Assignment – 5

Topics Covered: Definition, testing of single population means (t-test), difference of means: independent samples (t-test), paired samples t-test, F-test, definition, independence of attributes, introduction, assumptions, classification of ANOVA, one-way ANOVA.

A1. The purpose of a t-test is to determine if there is a significant difference between the ……… of two groups.

A2. What is the null hypothesis in a t-test?

A3. An F-test is typically used to compare the ……… of two or more groups to determine if they differ significantly.

A4. What does ANOVA (Analysis of Variance) help us test?

A5. In a paired samples t-test, each observation in one group is ………with an observation in the other group.

A6. What is the null hypothesis in a paired samples t-test?

A7. The alternative hypothesis in a paired samples t-test states that the mean difference between paired observations is ……….

A8. What is the degrees of freedom in a paired samples t-test with n number of pairs.

A9. To test independence of attributes, the observed data are arranged in a ………. table.

A10. How is the F-ratio calculated in an F-test?

B1. The heights of 10 adult males selected at random from a given locality had a mean of 158 cm and variance 39.0625 cm. Test at 5% level of significance, the hypothesis that the adult males of the given locality are on the average less than 162.5 cm tall.

B2. Two types of batteries, X and Y are tested for their length of life and the following results are obtained:

Battery	Sample size	Mean (hours)	Variance (hours)
A	10	1000	100
B	12	2000	121

Is there a significant difference between the two means?

B3. A travel agency's marketing brochure indicates that the standard deviations of hotel rates for two cities are the same. A random sample of 13 rooms rates in a city has a standard deviation of $ 27.50 and a random sample of 16 hotel room rates in the other city has a standard deviation of $ 29.75. Can you reject the agency's claim at $\alpha = 0.01$?

B4. In a certain sample of 2,000 families, 1,400 people are consumers of tea. Out of 1,800 Hindu families, 1,236 families are consumers of tea. State whether there is any significant difference between consumption of tea among Hindu and non-Hindu families.

B5. Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data:

Food I	Food II	Food III
8	4	11
12	5	8
19	4	7
8	6	13
6	9	7
11	7	9

C1. Certain pesticides are packed into bags by a machine. A random sample of 10 bags is drawn and their contents are found to weigh in kgs as follows: 50, 49, 44, 52, 45, 48, 46, 45, 49, 45 Test if the average packing can be taken as 50 kg.

C2. In a test given to two groups of students, the marks obtained are as follows:

First Group	18	20	36	50	49	36	34	49	41
Second Group	29	28	26	35	30	44	46		

Test if the average marks secured by two groups of students differ significantly.

C3. Two random samples were drawn from two normal populations and their values are as follows:

A	66	67	75	76	82	84	88	90	92		
B	64	66	74	78	82	85	87	92	93	95	97

Test whether the two populations have the same variance at 10% level of significance.

C4. 1,000 families were selected at random in a city to test the belief that the high income group families usually send their children to public schools and the low income families often send their children to government schools. The following results were obtained:

Income	Public School	Government School	Total
Low	370	430	800
High	130	70	200
Total	500	500	1000

Test whether income and type of schooling are independent.

C5. Three different techniques, namely medication, exercises and special diet are randomly assigned to (individuals diagnosed with high blood pressure) lower blood pressure. After four weeks the reduction in each person's blood pressure is recorded. Test at 5% level, whether there is significant difference in mean reduction of blood pressure among the three techniques.

Medication	10	12	9	15	13
Exercise	6	8	3	0	2
Diet	5	9	12	8	4

D1. The lifetime of electric bulbs for a random sample of 10 from a large consignment gave the following data: Life in thousand hours: 4.2, 4.6, 4.1, 3.9, 5.2, 3.8, 3.9, 4.3, 4.4, 5.6. Can we accept the hypothesis that the average lifetime of bulbs is 4000 hours?

D2. For a random sample of 10 persons fed on diet A the increased weight in pounds in a certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15, and 9. For another random sample of 12 persons fed on diet B, the increase in the same period were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10, and 17. Test whether the diets A and B differ significantly as regards their effect on increase in weight.

D3. Consider the following measurements of the heat producing capacity of the coal produced by two mines (in millions of calories per ton):

Mine 1	8260	8130	8350	8070	8340	
Mine 2	7950	7890	7900	8140	7920	7840

Can it be concluded that the two population variances are equal?

D5. Three composition instructors recorded the number of spelling errors which their students made on a research paper. At 1% level of significance test whether there is significant difference in the average number of errors in the three classes of students.

Instructor 1	2	3	5	0	8		
Instructor 2	4	6	8	4	9	0	2
Instructor 3	5	2	3	2	3	3	